

# Tight-BDT Lower-Bound Crosscheck — Leakage-Corrected ABCD

## Purity

Nominal parametric tight cut vs. flat BDT > 0.70/0.80/0.85/0.90/0.95

PPG12 internal note

2026-06-01

## Summary

To probe the sensitivity of the ABCD leakage-corrected purity to the tight-region BDT lower bound, the flat tight-cut crosscheck series (BDT > 0.70/0.80/0.85/0.90/0.95) was built out by adding the BDT > 0.80 and BDT > 0.70 points and running them through the full per-sample TTree pipeline. The leakage-corrected purity rises monotonically with the tight cut across the full range, exactly as expected. The nominal parametric cut is equivalent to a flat  $\approx 0.80$  threshold, and the loosest BDT > 0.70 point sits below the nominal at every bin. The comparison is robust. Two pre-existing config-hygiene issues in the flat-variant series were found, and neither biases this comparison.

## 1 Method

**ABCD regions.** The four regions are  $A$  = tight & isolated (signal region),  $B$  = tight & non-isolated,  $C$  = non-tight & isolated, and  $D$  = non-tight & non-isolated. The combinatorial background in  $A$  is estimated as  $R(BC)/D$ . The leakage-corrected purity is

$$P_{\text{leak}} = \frac{N_{\text{sig}}^A}{N_A},$$

solved self-consistently with the signal-MC leakage fractions

$$c_B, c_C, c_D = \frac{\text{signal MC in } B, C, D}{\text{signal MC in } A},$$

as implemented in `efficiencytool/CalculatePhotonYield.C`. The result is stored as the `TGraphErrors` `gpurity_leak` in `Photon_final_{var}.root`.

**Nominal tight cut.** The nominal tight boundary is parametric in transverse energy,

$$b_{\text{min}}(E_T) = 0.815625 - 0.0015625 \cdot E_T,$$

i.e. about 0.80 at  $E_T = 10$  GeV and 0.76 at  $E_T = 36$  GeV.

**Flat crosscheck variants.** Each flat variant sets the tight region to BDT >  $T$  (flat in  $E_T$ ) and the non-tight upper bound to  $T$  (no gap at the tight boundary), for  $T = 0.70, 0.80, 0.85, 0.90, 0.95$ .

**Pipeline.** All variants were produced with the full per-sample TTree-reader flow. Phase 1 (HT-Condor) ran `RecoEffCalculator_TTreeReader.C` on all 8 SI/DI MC sample pairs plus data per crossing-angle period, followed by `MergeSim` and per-period `CalculatePhotonYield`. Phase 2 merged the periods (lumi-weighted hadd) and ran the all-range `CalculatePhotonYield`. All results are all-range, all- $z$  nominal ( $64.3718 \text{ pb}^{-1}$ ), double-interaction-blended (0.224 at 0 mrad, 0.079 at 1.5 mrad), with truth-vertex reweighting on.

**Provenance.** The  $\text{BDT} > 0.85/0.90/0.95$  points already existed (produced 2026-05-14) and were verified current and sane, then reused. The  $\text{BDT} > 0.80$  (`flat_t80_nt50`) and  $\text{BDT} > 0.70$  (`flat_t70_nt50`) points were newly added: each config was mirrored from the `t85` config, changing only `var_type` and the two flat-cut intercepts ( $0.85 \rightarrow 0.80$  and  $0.85 \rightarrow 0.70$ ), and run end-to-end on 2026-05-31 and 2026-06-01 respectively.

## 2 Results

Table 1: Leakage-corrected purity (`gpurity_leak`) per  $E_T^\gamma$  bin center [GeV], all-range  $p+p$  at  $\sqrt{s} = 200$  GeV,  $64.37 \text{ pb}^{-1}$ . The  $\text{BDT} > 0.95$  series exhausts its last bin at  $E_T = 34$  GeV (entry shown empty).

| $E_T$ [GeV] | nominal | $\text{BDT} > 0.70$ | $\text{BDT} > 0.80$ | $\text{BDT} > 0.85$ | $\text{BDT} > 0.90$ | $\text{BDT} > 0.95$ |
|-------------|---------|---------------------|---------------------|---------------------|---------------------|---------------------|
| 11          | 0.511   | 0.473               | 0.512               | 0.541               | 0.583               | 0.664               |
| 13          | 0.563   | 0.527               | 0.559               | 0.594               | 0.655               | 0.774               |
| 15          | 0.624   | 0.586               | 0.614               | 0.647               | 0.695               | 0.801               |
| 17          | 0.727   | 0.691               | 0.710               | 0.740               | 0.768               | 0.866               |
| 19          | 0.697   | 0.666               | 0.671               | 0.729               | 0.798               | 0.930               |
| 21          | 0.656   | 0.674               | 0.669               | 0.671               | 0.744               | 0.870               |
| 23          | 0.871   | 0.848               | 0.886               | 0.929               | 0.963               | 1.000               |
| 25          | 0.877   | 0.817               | 0.819               | 0.876               | 0.912               | 0.787               |
| 27          | 0.828   | 0.634               | 0.889               | 0.835               | 0.879               | 0.924               |
| 30          | 0.730   | 0.652               | 0.630               | 0.607               | 0.743               | 0.947               |
| 34          | 0.983   | 0.972               | 0.960               | 0.992               | 1.010               | —                   |

Table 2: Region-A (tight & isolated) data yield, physical-bin integral, all-range, ordered loose to tight.

| selection           | region-A yield |
|---------------------|----------------|
| $\text{BDT} > 0.70$ | 96039          |
| nominal             | 84156          |
| $\text{BDT} > 0.80$ | 83625          |
| $\text{BDT} > 0.85$ | 75182          |
| $\text{BDT} > 0.90$ | 62744          |
| $\text{BDT} > 0.95$ | 37464          |

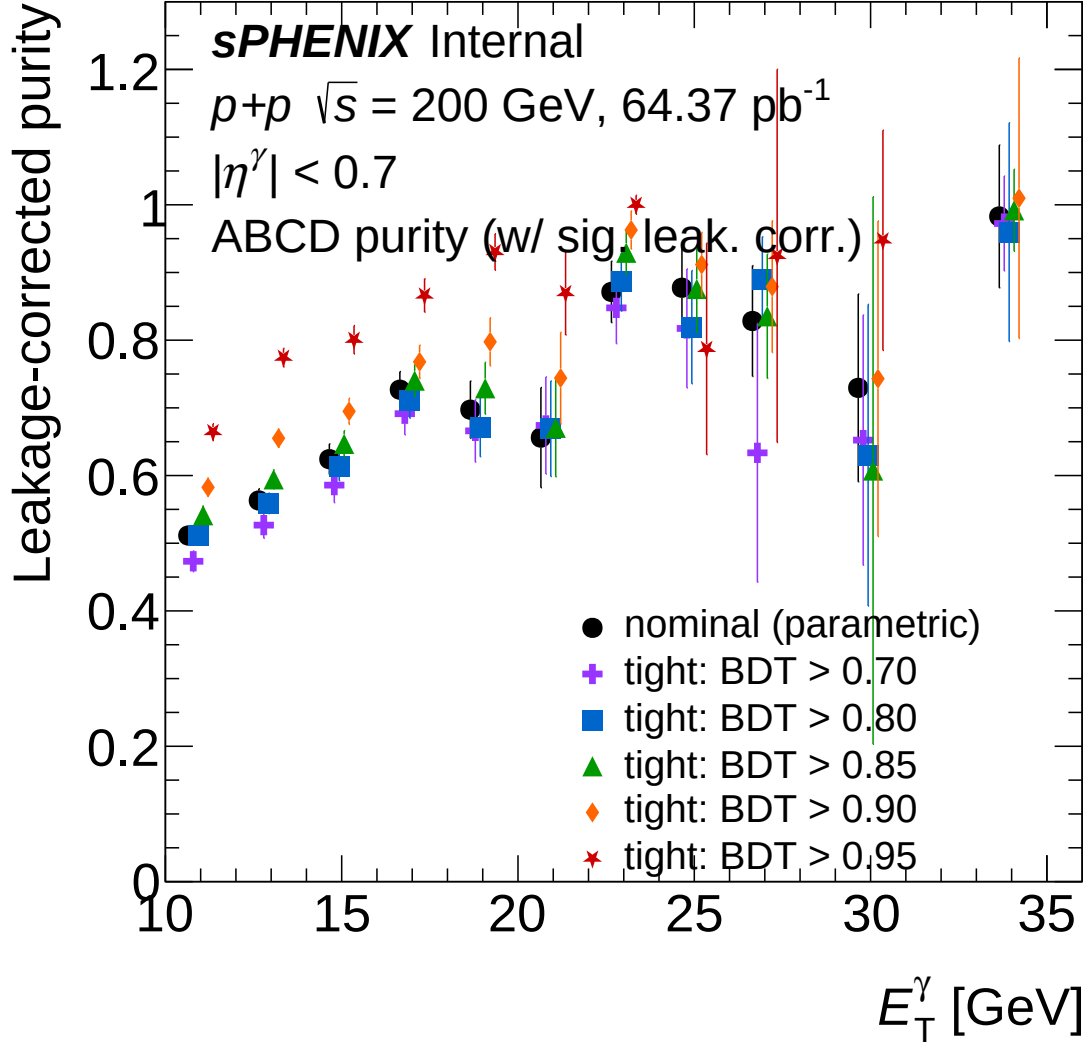


Figure 1: Leakage-corrected ABCD purity vs.  $E_T^\gamma$  for the nominal parametric tight-BDT cut and the flat tight cuts BDT > 0.70/0.80/0.85/0.90/0.95, all-range  $p+p$  at  $\sqrt{s} = 200 \text{ GeV}$ ,  $64.37 \text{ pb}^{-1}$ . Points are offset horizontally by up to 0.35 GeV for clarity. No smooth-fit overlay is drawn (points with vertical errors only).

#### Reading the result.

- **Monotonic rise with the tight cut.** The purity increases with the tight cut at every well-measured (low-to-mid  $E_T^\gamma$ ) bin, across the full range BDT > 0.70  $\rightarrow$  0.95. For example at  $E_T = 15 \text{ GeV}$  the purity goes  $0.59 \rightarrow 0.61 \rightarrow 0.65 \rightarrow 0.70 \rightarrow 0.80$  for BDT > 0.70/0.80/0.85/0.90/0.95. The loosest cut 0.70 sits below the nominal at every bin, and the tightest 0.95 gains roughly +0.15 to +0.25 absolute purity over the nominal/0.80 at low-mid  $E_T^\gamma$ .
- **Steep statistics-vs-purity trade.** Tightening from the loosest 0.70 to 0.95 discards a factor  $\sim 2.6$  of region  $A$  (96k  $\rightarrow$  37k clusters, Table 2), which is why the 0.95 series is noisy at high  $E_T^\gamma$ .
- **Nominal  $\equiv$  flat  $\approx 0.80$ .** The BDT > 0.80 variant has nearly the same region- $A$  yield as nominal (83625 vs. 84156, 0.6% lower) and tracks the nominal purity within the toy uncertainties. The 0.6% region- $A$  yield difference reflects the tight-boundary shape (nominal parametric, slightly looser than flat 0.80 at high  $E_T^\gamma$ ), and the small residual *purity* differences reflect the flat no-gap non-tight topology vs. the nominal tight/non-tight gap.
- **High- $E_T^\gamma$  is statistics-limited.** Above  $\sim 24 \text{ GeV}$  the comparison is dominated by toy un-

certainties (large error bars, occasional ABCD overshoot above 1). The  $\text{BDT} > 0.95$  series exhausts its last bin at  $E_T = 34 \text{ GeV}$  (shown empty). The figure shows points with vertical errors only, with no smooth-fit overlay.

### 3 Findings and caveats

Table 3: Findings from the flat-variant series. None biases the tight-cut purity comparison reported here.

| severity       | finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WARNING</b> | <b>nt_bdt_min override is shadowed.</b> The flat configs set <code>non_tight.bdt_min</code> (the flat non-tight floor labeled “nt50” = 0.50), but <code>RecoEffCalculator_TTreeReader.C</code> line 572 reads the inherited parametric <code>non_tight.bdt_min_intercept</code> (0.7333) $- 0.0133 \cdot E_T$ , which takes precedence. So the non-tight floor is the nominal parametric value in every flat variant, never the labeled flat 0.50. Consequence: <code>flat_t90_nt50</code> and <code>flat_t90_nt70</code> are numerically identical (verified: region <i>A, B, C, D</i> yields and <code>gpurity_leak</code> match to all digits). This does <i>not</i> bias the tight-cut purity comparison here, because the shadowed non-tight floor is identical (common-mode) across all of $\text{BDT} > 0.70/0.80/0.85/0.90/0.95$ and cannot drive the trend. The intended per-variant change is the shared tight floor and non-tight ceiling at $T$ . It only makes the “nt50/nt70” labels misleading. <i>Recommendation:</i> if a genuine flat non-tight floor is wanted, the variant generator must also set <code>non_tight.bdt_min_slope = 0</code> and <code>non_tight.bdt_min_intercept</code> to the flat value, not just <code>non_tight.bdt_min</code> . |
| <b>INFO</b>    | <b><math>\text{BDT} &gt; 0.95</math> Padé purity fit degenerates</b> on the near-unity points at high $E_T^\gamma$ (the stored <code>f_purity_leak_fit</code> is unreliable for 0.95). The crosscheck figure uses points only, so this does not affect it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>INFO</b>    | <b><math>\text{BDT} &gt; 0.80 \approx \text{nominal}</math>, <math>\text{BDT} &gt; 0.70</math> below it.</b> The flat 0.80 cut equals the nominal parametric cut to better than 1% in region- <i>A</i> yield, and the flat 0.70 cut is looser than nominal everywhere (region- <i>A</i> yield 96039, the largest of the series).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Reviewer pass matrix

| Artifact                                            | 2a physics                                                     | 2b cosmet-<br>ics       | 2c re-derivation                                             | 3.5 fix-validator                                                        |
|-----------------------------------------------------|----------------------------------------------------------------|-------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>flat_t80/flat_t70</code><br>config + pipeline | PASS (comparison<br>valid, <code>nt_min</code> bug<br>flagged) | N/A                     | PASS (t80, t70 pu-<br>rity + <i>A</i> -count con-<br>firmed) | PASS (Phase<br>1+2 <code>rc=0</code> ,<br><i>A</i> =83625/96039<br>sane) |
| purity comparison<br>figure                         | N/A                                                            | PASS (de-<br>ploy gate) | PASS                                                         | N/A                                                                      |
| existing<br><code>t85/t90/t95</code>                | PASS                                                           | shown in<br>figure      | PASS (all con-<br>firmed)                                    | N/A                                                                      |

### Files

New.

- `efficiencytool/config_bdt_flat_t{80,70}_nt50{,_0rad,_1p5mrad}.yaml`
- `efficiencytool/flat_t{80,70}_feeders.list`, `flat_t{80,70}_phase1.sub`
- `plotting/plot_purity_tightbdt_crosscheck.C`
- `plotting/figures/purity_tightbdt_crosscheck.pdf`
- `efficiencytool/results/Photon_final_bdt_flat_t{80,70}_nt50{,_0rad,_1p5mrad}{,_mc}.root`

**Modified.**

- `efficiencytool/make_bdt_variations.py` (added `flat_t80_nt50` and `flat_t70_nt50` variants)
- `efficiencytool/all_feeder_configs.list` and `all_range_configs.list` (added `t80`, `t70` entries)